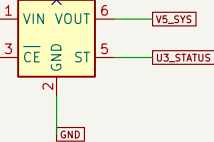
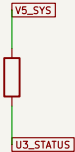


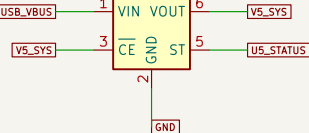
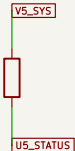
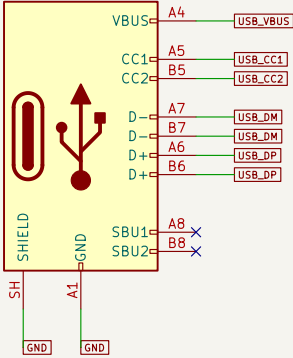
	1	2	3	4	5	6	7	8
A	Integrated draft: power chain (sheets 2–4) and board stage (sheets 5–8). See DESIGN_SPEC.md. Rails: V5_SYS from the 5 V inputs, V3V3_SYS from the TPS62913 buck, V3V3_ANA from the TPS7A20 LDO. No PCB placement or routing has been performed.							
B	POWER 1 / 5 V in, USB, source isolation File: power_sources.kicad_sch				BOARD 2 / IMU, barometer, high-g File: sensors.kicad_sch			
C	POWER 2 / 3.3 V system buck and analog LDO File: power_3v3.kicad_sch				BOARD 3 / log flash and microSD File: storage.kicad_sch			
D	POWER 3 / peripheral ports File: power_periph.kicad_sch				BOARD 4 / PWM outputs File: actuators.kicad_sch			
E	BOARD 1 / RP2354B File: mcu.kicad_sch							
F	1	2	3	4	5	6	7	8

Sheet: / File: MARV-V2.kicad_sch		
Title: MARV integrated power – preliminary		
Size: A3	Date: 2026-09-14	Rev: P0 – NOT FOR FAB
KiCad E.D.A. 10.0.6	Id: 1/8	

B



C



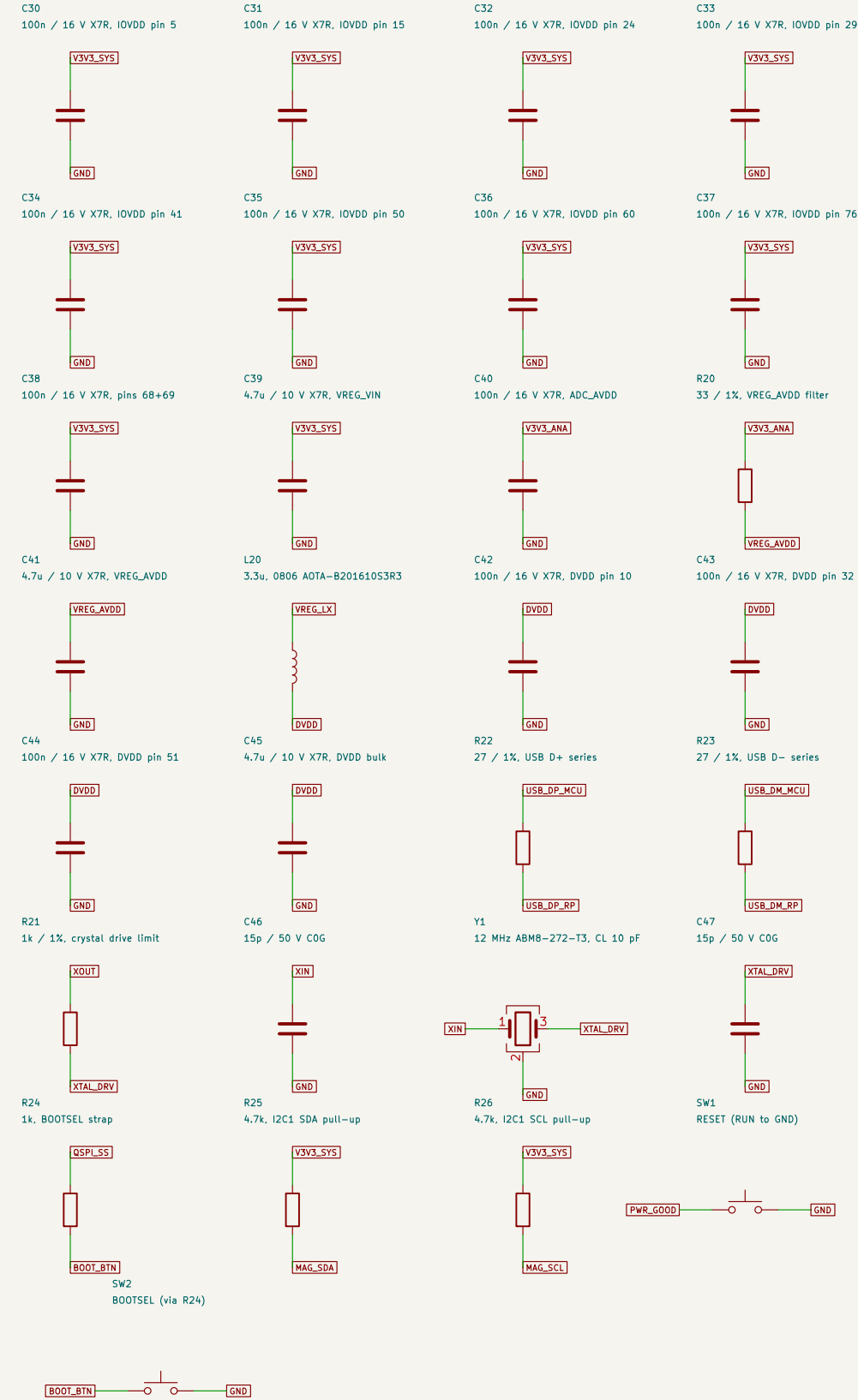
E

AVIONICS ONLY: 300 mA continuous / 500 mA short peak, provisional. NO SERVO POWER.





RP2354B, QFN-80, 2 MB in-package QSPI flash. Digital rails on V3V3_SYS, analog rails on V3V3_ANA, core DVDD from the on-chip buck. Decoupling, core regulator, crystal and BOOTSEL follow the RP2350 hardware design guide; every pin net is a global label.



Crystal circuit per design guide Sec 4: ABM8-272-T3, 15 pF each side (7.5 pF series + ~3 pF stray = 10.5 pF vs CL 10 pF), 1 k in the XOUT leg to limit drive with a 50 ohm ESR crystal. C38 is the single design-guide decoupling exception (pins 68+69 share it). VREG_AVDD RC = 33 ohm + 4.7 uF (design guide Sec 2.1); VREG_FB is tied to DVDD, VREG_PGND to GND, EP (pin 81) to GND.

Telemetry dividers: 10k/15k gives 3.00 V at 5.00 V in, 3.15 V at 5.25 V; 100 nF at each ADC pin. Buzzer runs from V5_SYS (avionics OR output), never from the servo bus; D22 clamps the coil kick. QSPI_* are the dedicated QSPI pads; they reach the in-package flash die AND the package pins (RP2350 datasheet Sec 14.3), so U24 shares the bus with GPIO0/QMI_CS1n (FLASH_CS1) as its select. RP2354 requires QSPI_IOVDD = 3.3 V, and IOVDD = 3.3 V to run a second QSPI device (Sec 14.9).

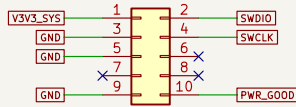
U20
RP2354B
#FLG5
PWR_FLAG
#FLG6
PWR_FLAG
#FLG7
PWR_FLAG

PORT NAMING: a port label is named for the peripheral pin it comes from, so GPS_TX / ELRS_TX are module outputs and land on MCU UART RX pins, while GPS_RX / ELRS_RX are module inputs fed by the MCU UART TX pins. J6/J7/J9 (POWER 3) are 3.3 V logic powered directly from V3V3_SYS.



GPIO MAP (pin = QFN-80 pin)
GPIO0 77 FLASH_CS1 QMI CS1n, log flash select
GPIO1 78 (no connect) spare
GPIO2 79 PWM1 PWM1 A
GPIO3 80 PWM2 PWM1 B
GPIO4 1 PWM3 PWM2 A
GPIO5 2 PWM4 PWM2 B
GPIO6 3 MAG_SDA I2C1 SDA
GPIO7 4 MAG_SCL I2C1 SCL
GPIO8 6 ELRS_RX UART1 TX -> radio RX
GPIO9 7 ELRS_TX UART1 RX <- radio TX
GPIO10 8 PWM5 PWM5 A
GPIO11 9 PWM6 PWM5 B
GPIO12 11 GPS_RX UART0 TX -> GPS RX
GPIO13 12 GPS_TX UART0 RX <- GPS TX
GPIO14 13 PWM7 PWM7 A
GPIO15 14 PWM8 PWM7 B
GPIO16 16 SENS_MISO SPI0 RX
GPIO17 17 HG_ACC_CS ADXL375 CS
GPIO18 18 SENS_SCK SPI0 SCK
GPIO19 19 SENS_MOSI SPI0 TX
GPIO20 20 IMU_ACC_CS BMI088 CSB1
GPIO21 21 IMU_GYR_CS BMI088 CSB2
GPIO22 22 BARO_CS BMP581 CSB
GPIO23 23 IMU_ACC_INT BMI088 INT1

J10
SWD, Cortex-M 10-pin 1.27 mm



J10 pin 1 is 3V3 OUT for debugger reference only – never power the board from it. Pin 10 is the debugger reset line on PWR_GOOD/RUN: pins 6-8 (SWD, KEY, NC) unused. RUN is held by the TPS62913 power-good pull-up on POWER 2, so the MCU cannot run before V3V3_SYS is in regulation.

R27
10k / 1%, 5V_IN top

R30
15k / 1%, VBUS bottom

D20
green, status

R33
1k, gate series

R34
100k, gate pull-down

Q20
2N7002

BZ1
5 V active buzzer

D22
flyback across BZ1

V5_SYS

BUZZ_D

BUZZ_G

BUZZ_G

BUZZ_G

BUZZ_G

BUZZ_G

BUZZ_G

BUZZ_G

BUZZ_G

BUZZ_G

BUZZ_G

BUZZ_G

BUZZ_G

R28
15k / 1%, 5V_IN bottom

C49
100n / 16 V X7R, ADC1

VBUS_SENSE

VBUS_SENSE

VBUS_SENSE

VBUS_SENSE

VBUS_SENSE

VBUS_SENSE

VBUS_SENSE

VBUS_SENSE

VBUS_SENSE

VBUS_SENSE

VBUS_SENSE

VBUS_SENSE

VBUS_SENSE

VBUS_SENSE

VBUS_SENSE

VBUS_SENSE

VBUS_SENSE

VBUS_SENSE

VBUS_SENSE

VBUS_SENSE

C48
100n / 16 V X7R, ADC0

VIN_SENSE

VIN_SENSE

VIN_SENSE

VIN_SENSE

VIN_SENSE

VIN_SENSE

VIN_SENSE

VIN_SENSE

VIN_SENSE

VIN_SENSE

VIN_SENSE

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VIN_SENSE

VIN_SENSE

VIN_SENSE

VIN_SENSE

VIN_SENSE

VIN_SENSE

R29
10k / 1%, VBUS top

VBUS_SENSE

VBUS_SENSE

VBUS_SENSE

VBUS_SENSE

VBUS_SENSE

VBUS_SENSE

VBUS_SENSE

VBUS_SENSE

VBUS_SENSE

VBUS_SENSE

VBUS_SENSE

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VBUS_SENSE

VBUS_SENSE

VBUS_SENSE

VBUS_SENSE

VBUS_SENSE

VBUS_SENSE

VBUS_SENSE

VBUS_SENSE

LED1_A

LED2_A

LED1_A

LED2_A

LED1_A

LED2_A

LED1_A

LED2_A

LED1_A

LED2_A

LED1_A

LED2_A

LED1_A

LED2_A

LED1_A

LED2_A

LED1_A

LED2_A

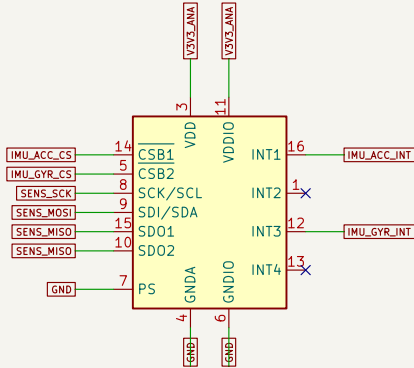
LED1_A

LED2_A

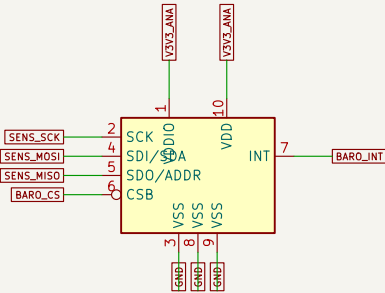
LED1_A

LED2_A

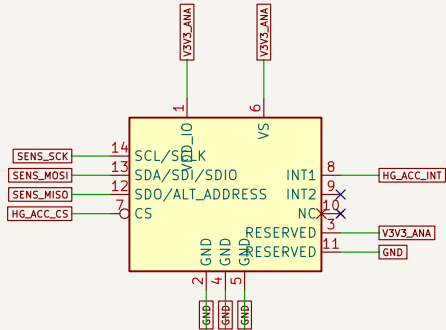
All three sensors run 4-wire SPI on the shared SENS_SCK / SENS_MOSI / SENS_MISO bus with one chip select each, and are powered from V3V3_ANA (TPS7A20 LDO) with 100 nF + 1 uF at every supply pin per datasheet.



U21
BMI088 6-axis IMU



U22
BMP581 barometer



U23
ADXL375 high-g accelerometer

C50
100n / 16 V X7R, U21 VDD



C51
1u / 10 V X7R, U21 VDD



C52
100n / 16 V X7R, U21 VDDIO



C53
1u / 10 V X7R, U21 VDDIO



C54
100n / 16 V X7R, U22 VDD



C55
1u / 10 V X7R, U22 VDD



C56
100n / 16 V X7R, U22 VDDIO



C57
1u / 10 V X7R, U22 VDDIO



C58
100n / 16 V X7R, U23 VS



C59
1u / 10 V X7R, U23 VS



C60
100n / 16 V X7R, U23 VDD_IO



C61
1u / 10 V X7R, U23 VDD_IO



U21 BMI088 (datasheet Sec 6.2): PS tied to GND selects SPI. CSB1 = accelerometer (IMU_ACC_CS). CSB2 = gyroscope (IMU_GYR_CS). SD01 is the accelerometer data output and SD02 the gyroscope one, both high-Z while deselected, so they share SENS_MISO (the symbol is Sensor_Motion:BMIO88 with SD02 retyped passive so ERC accepts that). INT1 = accelerometer interrupt, INT3 = gyroscope interrupt; INT2 and INT4 are unused and left open. VDD and VDDIO are both on V3V3_ANA; GND and GNDIO are the same board ground.

U22 BMP581: CSB = BARO_CS, SDI = SENS_MOSI, SDO = SENS_MISO, SCK = SENS_SCK, INT = BARO_INT. VDDIO and VDD on V3V3_ANA, all three VSS pins to GND.
U23 ADXL375: CS = HG_ACC_CS, SDA/SDI = SENS_MOSI, SDO = SENS_MISO, SCL/SCLK = SENS_SCK, INT1 = HG_ACC_INT. Per the pin function table, pin 3 (RESERVED) goes to VS and pin 11 (RESERVED) to GND; pin 10 (NC) and the unused INT2 are left open. VS and VDD I/O are both on V3V3_ANA.

Sheet: /BOARD 2 / IMU, barometer, high-g/
File: sensors.kicad_sch

Title: BOARD 2 / IMU, barometer, high-g

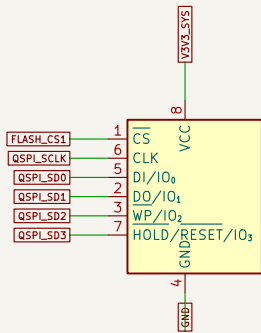
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-14

Rev: P0 - REVIEW ONLY

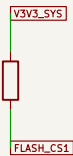
Id: 6/8

Log flash on the shared QSPI bus with GPIO0/QMI CS1n as its chip select, and a latched microSD wired for 4-bit PIO (also SPI-compatible). Both on V3V3_SYS with local bulk, per the rail split in DESIGN_SPEC.



U24
W25Q128JVSIQ, 128 Mbit QSPI NOR

R35
10k, FLASH_CS1 pull-up



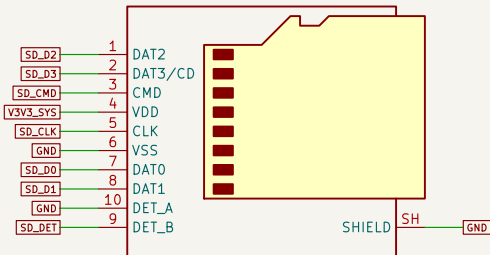
C62
100n / 16 V X7R, U24 VCC



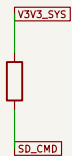
C63
1u / 10 V X7R, U24 VCC



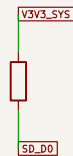
J11
microSD, Molex 104031-0811 push-push



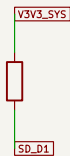
R36
10k, SD_CMD pull-up



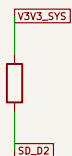
R37
10k, SD_D0 pull-up



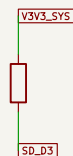
R38
10k, SD_D1 pull-up



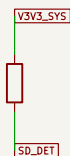
R39
10k, SD_D2 pull-up



R40
10k, SD_D3 pull-up



R41
10k, SD_DET pull-up



C64
100n / 16 V X7R, at the socket



C65
10u / 10 V X7R, at the socket



C66
47u / 6.3 V X5R, at the socket



U24 shares the dedicated QSPI pads with the RP2354B in-package flash die (datasheet Sec 14.3): CLK = QSPLSCLK, DI/IO0 = QSPLSD0, DO/IO1 = QSPLSD1, WP/IO2 = QSPLSD2, HOLD/IO3 = QSPLSD3. Only the chip select differs – FLASH_CS1 comes from GPIO0 (QMI CS1n), pulled up to V3V3_SYS by R35 so the part is deselected before the MCU drives it. QSPI_SS (the internal die) keeps its own strap on the MCU sheet.

J11: DET_A to GND and DET_B to SD_DET with a 10k pull-up, so the input reads low with a card inserted. The shield goes to GND. 100 nF + 10 uF + 47 uF sit at the socket (DESIGN_SPEC "Rails": SD transients stay on the buck rail), 10k pull-ups on CMD and DAT0–DAT3 are the SD-standard idle bias and also keep DAT3/CD high for 4-bit mode. VDD = V3V3_SYS.

Sheet: /BOARD 3 / log flash and microSD/
File: storage.kicad_sch

Title: BOARD 3 / log flash and microSD

Size: A3
KiCad E.D.A. 10.0.6

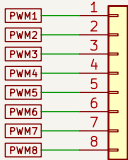
Date: 2026-09-14

Rev: P0 – REVIEW ONLY

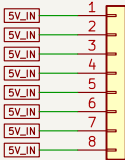
Id: 7/8

Eight PWM outputs on a 3-row 2.54 mm block: signal /2SERVO_5V / GND. No series resistors –3.3 V logic straight from the RP2354B, which is what servos and ESCs expect.

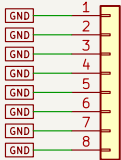
J12
PWM signals, row 1



J13
SERVO_5V pass-through, row 2



J14
GND, row 3



3.3 V logic; servo/ESC power is the external BEC bus, 3 A limit is the BEC's.

Row 2 is SERVO_5V = the 5V_IN net, i.e. the J3 input BEFORE U3 (the LM66100 ideal diode). Servo and ESC current therefore never crosses the board rails, V5_SYS or the OR-ing devices; it returns through the row 3 ground pins. Nothing on this sheet loads V3V3_SYS or USB_VBUS.

No 3-row generic connector symbol or 2.54 mm 3x08 footprint ships with this KiCad install, so the header is drawn as three stacked Conn_01x08 symbols (J12 signal, J13 SERVO_5V, J14 GND) with three 1x08 footprints. Place them on 2.54 mm centres to form the usual 3x8 servo block; column n is J12 pin n / J13 pin n / J14 pin n.

PWM MAP
PWM1 = GPIO2 PWM5 = GPIO10
PWM2 = GPIO3 PWM6 = GPIO11
PWM3 = GPIO4 PWM7 = GPIO14
PWM4 = GPIO5 PWM8 = GPIO15

Sheet: /BOARD 4 / PWM outputs/
File: actuators.kicad_sch

Title: BOARD 4 / PWM outputs

Size: A3 Date: 2026-09-14
KiCad E.D.A. 10.0.6

Rev: P0 – REVIEW ONLY
Id: 8/8